Learning Objectives

- Be able to
  - Integrate location decisions with operations strategy and relate them to globalization
  - Discuss several network choices
    - Centralized or distributed
    - Standardized or localized
    - Separated or integrated
  - Apply four types of location analysis
    1. Geographic or spatial analysis and TLC
    2. Network analysis
    3. Competitive analysis
    4. Facility role or focus analysis

What drives capacity location and globalization?

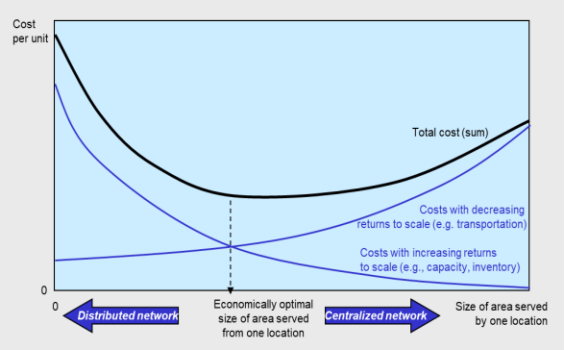
- Competitive Strategy
  - Product/Service positioning
  - Competitors' positions
- Product Markets
  - Predicted growth, uncertainty, and geographical distribution of demand
  - Customer proximity premium
  - Distinctiveness of local product markets
- Supply Markets
  - Availability and costs of inputs (capital, labor, energy, materials, goods, services …)
- Technology
  - Production economies (or diseconomies) of scale, scope and learning
  - Transportation costs and lead times
- Macro-economic and non-market factors
  - Tariffs, duties, quotas, and trade agreements
  - Tax rates and government incentives
  - Exchange rates and volatility
  - IP protection
  - Political stability
- Existing resources and processes
  - Existing facility network
  - Existing skills and capabilities

Integrate Capacity Location with operations Strategy

- Location decisions are intimately related to an organization's operations strategy and must be integrated with competitive strategy, e.g.,
  - Cost reduction -> locate near source of resources
  - Convenience for customers -> many locations and in high 'traffic' areas
- Location decisions impact (and vice versa)
  - Competencies: costs, responsiveness (LT), quality, flexibility
    - Capacity Size: investment and capital, taxes and incentives, storage and inventory
    - Capacity Timing: LT to start up, ramp-up/down time
    - Capacity Type: degree of flexibility and automation
    - Supply and Demand: labor factors, suppliers and integration, supply chain configuration, and the market demand factors, macroeconomic and non-market factors
    - Innovation and technology – availability of knowledge and skills, state of infrastructure

Network choices

- Should a network be centralized or distributed?
  - A distributed network offers customer proximity and basis for differentiation
  - Centralization is typically observed when “bigger is better”, risk pooling stems from aggregating volume and volatilities, sometimes can better guarantee consistence and uniform quality
  - Costs:
    - Total network costs per unit includes a set of costs that decrease and another that increase with the size of the served area



Network choices

- Should network be separated or integrated?
- Separation leads to
  - market focused plants (local product and supply markets) with resulting operational efficiencies (valuable if customers are willing to pay for this)
  - plant networks that are easy to manage
- Integration (for distributed facilities) means that one location can serve multiple markets or multiple locations can serve one market
  - offers the scale economies of capacity pooling, less safety capacity is needed
  - comes in different levels of operational flexibility (level of integration)
    - consists of flexible capacity (to produce for multiple markets), contingent allocation of capacity to markets and transshipments as needed
  - allows reaction to demand and exchange rate shifts (demand and currency risk)
    - Integration becomes more valuable when demands and exchange rates become more volatile

Network choices

- Should network be standardized or localized?
  - Standardization
    - offers purchasing power for materials and equipment, consistent and uniform quality
    - facilitates integration and its associated advantages (e.g. risk pooling)
    - can slow down improvements (enforcing standards inhibits natural experimentation) but can also increase the impact of improvements (can be copied to other facilities)
  - Localization (for a distributed network)
    - may be required (by regulation)
    - may be useful to adjust facilities to available inputs/suppliers (e.g., Mittal Steel mini-mills; MBA Polymers plastic recycling plants)
    - can be part of a differentiation strategy that takes advantage of local customer preferences (e.g., Pirelli 30-inch tires in USA)

Decision guidelines

- Tailor the locations in your network
  - To fit your strategies and local conditions
- Think globally: consider costs and revenues
  - Globalization is certain to increase
  - Re-optimization of value chain by including offshore locations can reduce cost by up to 70% (McKinsey)
    - Cutting down costs allows companies to decrease price
    - price drop from $10000 to $7000 of a car can double the demand (Farrell 2004)
- Global network flexibility (degree of integration) is valuable

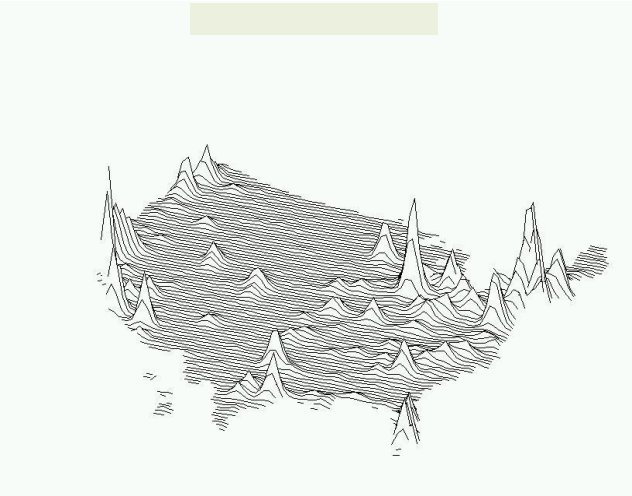
Four types of Location analysis

1. Geographical or Spatial analysis and Total Landed Costs
2. Network analysis
3. Competitive analysis
4. Facility role or focus analysis

1.Geographical or Spatial analysis

- Starts with estimates of market demand per region and of the number of needed potential facilities, and ends by comparing all potential sites' Total Landed Costs (TLC)
- Used when transportation costs constitute a significant fraction of total costs such as
  - Cement industry
  - Fiberglass insulation industry
  - Fresh and frozen food industry

<u>Example:</u> A global appliance manufacturer used a density map of its US customer base, along with other information, to make decisions about where to offer premium service operations.



http://nationalatlas.gov/

What's Total Landed Cost (TLC)?

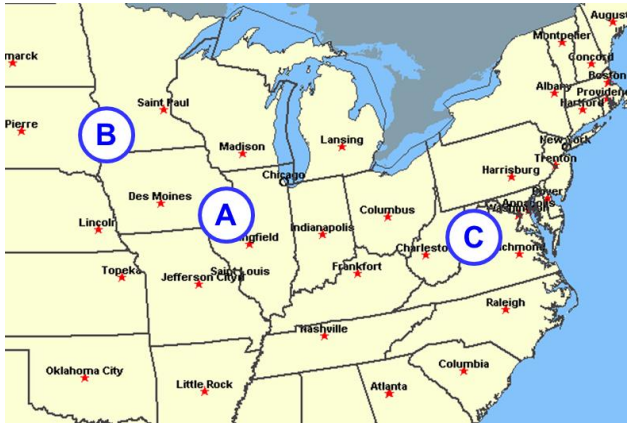
- Total end-to-end cost to transform inputs at sourcing location to outputs at customer locations. (Scope: products x customers)
- TLC has a cost focus
  - It is therefore important to fix the differentiation level on dimensions of time, quality, variety/customization and estimate what it would take to reach that same level of differentiation in different locations.
- The challenge is to carefully track direct and *indirect (often hidden)* costs through the entire value chain. This analysis is painful (because data is not always easy to find) but rewarding.
- It is some of the most quantitative studies in operations
  - Costs of inputs, transportation, labor, inventory … good analysis can save 5% on COGS

2. Network analysis

- Considers the entirety of a company's facility location and interconnections which can be used in conjunction with spatial analysis
- Can capture the interdependency with size and timing

Example

- You are starting operations in the US
- Assume you have a short-term view: build one plant in A
- If, however, you anticipate potential future additions: start in B so that in the longer term you add C, so that TLC(B+C) < TLC(A)



3. Competitive analysis

- Takes location of competitors or their possible reaction into account
  - e.g. competitors locate side by side in the middle of main street, such as car dealers
  - The co-location of competitors is in line with predications from game theory (Harold Hotelling, 1929)
- Important if customers' choice is based on proximity

4. Facility role or focus analysis

- Facility location strategy needs to specify the role and competences of a specific site
- To decide on the role, we must assign activities and responsibilities to individual facilities, e.g.,
  - electrics industry often separate and assign R&D, component manufacturing, assembly, and test to parent plants, component plants, assembly and test locations
  - Eli Lily separates bulk manufacturing and fff plants
- The strategic role of a facility location not only assigns activities but also clearly specifies responsibilities and competences as part of the (global) network
- To help articulate this strategic role, Ferdows (1997) suggests that managers start their facility focus analysis by answering the below two questions:
  1. What is the primary strategic reason for the facility's location?
  2. What is the scop of its current activities and competences?

Strategic role matrix for facilities

- By Ferdows (1997)

What is the scope of activities and competencies?

Broad & high

Source plant • Produces at low cost but with same ability as best plant in the network • Authority over procurement, Production, planning, logistics, process & product customization	**Lead plant** • Global hub for product and process knowledge and innovation. • Full authority over all activities. Taps into local skills and knowledge to initiate company-wide innovation.	**Contributor plant** • Serves specific regional market but competes with other plants in network for new processes or products. • Authority over product and process development as well as supplier choice.
Offshore plant • Produces specific items at low cost and exports for further work or for sale • Minimal authority and investment.	**Outpost plant** • Primary role is to collect information from advanced suppliers, competitors, research labs, or customers. • Secondary role as offshore or server.	**Server plant** • Serves specific regional market to overcome tariffs, taxes, logistics, or foreign-exchange risk. • Limited authority to make minor modifications to fit local conditions.

Narrow & low

Access to low-cost production | Access to skills and knowledge | Access to market

What is the primary strategic reason for the location?